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Systematic review

Efficacy of acupuncture in the treatment of osteoporosis-related fractures: A systematic review and Bayesian network meta-analysis

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ABSTRACT

Introduction: In recent years, there has been an increasing interest in the utilization of different acupuncture regimens as alternative therapies for fractures linked to osteoporosis. Nevertheless, the efficacy of these diverse treatment choices continues to be a subject of contention. To tackle this matter, a comprehensive Bayesian network meta-analysis has been undertaken to thoroughly assess the potential advantages of various acupuncture techniques in the management of osteoporosis-related fractures.

Methods: Until April 12, 2023, an extensive search was carried out to identify randomized controlled trials investigating the application of acupuncture in the treatment of fractures related to osteoporosis. The search encompassed multiple databases, such as PubMed, EMBASE, Web of Science, Cochrane Library, China Knowledge Network (CNKI), WanFang, VIP Database, and SinoMed. The network meta-analysis specifically aimed to evaluate the efficacy of different acupuncture techniques in conjunction with conservative biomedical treatment or invasive biomedical surgical treatment for managing fractures related to osteoporosis.

Results: A total of 35 articles were included, involving 2,727 patients. These articles examined and investigated five different types of acupuncture treatment options. The evaluated acupuncture treatments included ordinary acupuncture, electroacupuncture, floating acupuncture, needle-knife, and warm acupuncture. Results showed that when electroacupuncture was combined with daily care, it resulted in a significant decrease in both the Visual Analogue Scale (VAS) score and the Oswestry Disability Index (ODI) score. Specifically, the VAS score decreased by -2.29 (95 % CI: -3.84 ~ -0.75), while the ODI score decreased by -22.03 (95 % CI: -33.88 ~ -10.54) compared to those who received only daily care. Additionally, there was a significant decrease in the VAS score when comparing the combination of ordinary acupuncture with percutaneous vertebroplasty (PVP) surgery to PVP surgery alone. The VAS score decreased by -1.39 (95 % CI: -2.73 ~ -0.04) in the group that underwent both acupuncture and surgery. Similarly, in the comparison between warm acupuncture combined with percutaneous kyphoplasty (PKP) surgery and PKP surgery alone, there was a significant reduction in the VAS score. The VAS score decreased by -3.62 (95 % CI: -5.49 ~ -1.74) in the group that received both acupuncture and surgery.

Conclusions: The implementation of a well-balanced acupuncture treatment regimen can effectively alleviate pain and disability experienced by patients with fractures related to osteoporosis, irrespective of whether surgical intervention is conducted.

Abbreviations: PVP, percutaneous vertebroplasty; PKP, percutaneous kyphoplasty; NMA, network meta-analysis; VAS, Visual Analogue Scale; ODI, Oswestry Disability Index; BMD, bone mineral density; MCMC, Markov Chain Monte Carlo; DIC, Deviance information criterion.

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1. Introduction

Osteoporosis, a bone disease characterized by decreased bone density and impaired bone structure, often results in fractures in the spine, hip, and wrist among older individuals [1,2]. These fractures significantly contribute to disability and mortality rates among older individuals [3]. Research indicates that approximately one-third of women and one-fifth of men aged 50 and above will experience fractures related to osteoporosis during their lifetime [4]. As the population ages, the impact of these fractures on individuals' quality of life and the economic burden they place on families and society becomes increasingly substantial [5]. China is a highly populous country, and it is projected that the annual number of osteoporosis-related fractures will reach 4.6 million by 2035, with an estimated treatment cost of USD 20 billion. By 2050, these numbers are expected to rise to 6 million fractures annually, accompanied by USD 25.4 billion in treatment expenses [6]. Therefore, it is imperative to urgently discover effective treatment methods for osteoporosis-related fractures and alleviate symptoms. By doing so, the adverse effects of this disease on population health and healthcare budgets can be mitigated.

The current primary method for treating fractures caused by osteoporosis is through minimally invasive biomedical surgery and conservative biomedical treatment. Percutaneous vertebroplasty (PVP) and percutaneous kyphoplasty (PKP) are commonly employed minimally invasive surgical procedures for patients who have compression fractures in the thoracolumbar region [7]. Although surgical intervention has been reported to be effective in reducing the length of bed rest, relieving pain, and promoting fracture site stability [8,9], careful monitoring of contraindications is crucial due to the considerable financial burden on patients associated with the high cost of treatment. Additionally, it is important to note that postoperative persistent pain may adversely affect daily activities, and there is a potential risk of refractures subsequent to the surgical procedure [10–12]. There is a preference for biomedical conservative treatment among patients due to economic limitations. This treatment option includes bed rest, oral analgesics, calcium supplements, and external fixation at the fracture sites [13]. Research studies have found that both invasive biomedical surgery and biomedical conservative treatment offer similar long-term efficacy and have similar rates of refractures [14,15]. However, it is important to note that the duration of biomedical conservative treatment is longer. Typically, patients require approximately three months of treatment to experience significant pain reduction, restoration of limb function, and improvement in their quality of life [16].

For over 2000 years, acupuncture has been utilized as a non-drug therapy based on the ancient Chinese medical system's meridian theory. Its growing popularity can be attributed to its efficacy in treating a wide range of diseases, along with its minimal side effects [17,18]. Scientific studies have demonstrated that acupuncture is highly effective in relieving chronic pain and providing enduring relief [19,20]. Additionally, when used in conjunction with conventional biomedical treatments, acupuncture has been found to enhance limb function in post-stroke spasticity patients [21]. Acupuncture has been integrated into the management of osteoporosis-related fractures, yielding positive outcomes. With its roots in China and well-documented in the Huang Di Nei Jing, acupuncture encompasses a diverse range of nine needle types, each employing a distinct technique customized to address the patient's specific condition, ailment location, and body constitution. Over time, these needles have undergone refinements, leading to the development of various acupuncture methods widely utilized in modern clinical practice. These include ordinary acupuncture, three-edged acupuncture, needle-knife, electroacupuncture, intradermal acupuncture, warm acupuncture, floating acupuncture, wrist and ankle acupuncture, as well as head acupuncture [22]. Commonly used techniques for treating fractures related to osteoporosis include ordinary acupuncture, warm acupuncture, electroacupuncture, floating acupuncture, and needle-knife techniques. However, there is a paucity of comparative studies assessing the efficacy of these different acupuncture methods. As a result, there is ongoing debate among clinicians regarding the optimal selection of acupuncture techniques for combination treatment. This uncertainty leaves clinicians unsure about which acupuncture therapies to employ.

Network meta-analysis, which is an improvement on traditional bivariate analysis [23], enables the concurrent assessment of different acupuncture therapies, both directly and indirectly. It offers a method to evaluate the comparative efficacy of diverse acupuncture treatments through comprehensive comparison outcomes. By integrating direct evidence from within-study pairwise comparisons with indirect evidence from across-study control comparisons [24], network meta-analysis allows for the comparison of multiple treatments even in the absence of direct comparisons in previous studies, compared to traditional meta-analysis. Moreover, Bayesian-based network meta-analysis provides probability estimates that assist clinicians in making well-informed interventions and treatment decisions [25]. The Bayesian approach removes the need for assuming a normal distribution and is especially beneficial when each pairwise comparison involves a limited number of studies, as opposed to the frequentist method.

This review utilized Bayesian network meta-analysis on randomized controlled trials to compare and evaluate the efficacy of different acupuncture methods in combination with surgical or conservative treatment for fractures related to osteoporosis. The objective is to present patients with diverse treatment goals the optimal choice for combination treatment. The results intend to offer substantiated evidence to assist them in making informed decisions.

2. Methods

2.1. Review registration

The meta-analysis guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [26] were followed in the present review. Furthermore, the review has been registered on PROSPERO (https://www.crd.york.ac.uk/prospero/) under the registration number CRD42023426620.

2.2. Eligibility criteria

2.2.1. Inclusion criteria

P (Population): Fractures caused by primary osteoporosis were observed in patients, regardless of their age, gender, or the location of the fracture.

I (Intervention): This review aimed to evaluate the efficacy of different interventions in the treatment of the trial group. The interventions included acupuncture combined with surgical treatments such as PVP, PKP surgical treatment, or non-surgical conservative treatment. The acupuncture modalities used in this review comprised ordinary acupuncture, warm acupuncture, electroacupuncture, needle-knife, floating acupuncture, among others.

C (Comparison): The patients in the control group were randomly assigned to receive either surgical treatment or daily care. In order to mitigate any potential ambiguity arising from variations in the definition of conservative treatment between China and other countries, many Chinese researchers, doctors, and patients often include traditional Chinese medicine therapies such as acupuncture and biomedical conservative treatment within the conservative treatment category. However, other countries list acupuncture and other traditional Chinese medicine therapies separately. In clinical studies, doctors consistently advise patients with osteoporosis-related fractures, regardless of whether they receive surgical treatment or not, to supplement calcium, avoid certain activities, and rest in bed. Therefore, for the sake of clarity, this review collectively refers to daily care as conservative treatments received by patients with osteoporosis-related fractures, excluding acupuncture.

Table 1
Basic information of the included studies.

First author	Publication period	Author's country	Diagnostic criteria for osteoporotic vertebral compression fractures	Randomization method	Course of disease	Fracture site	Interventional protocol
Mingxing Lai [49]	2022	China	Refer to Guidelines for the Diagnosis and Treatment of Primary Osteoporosis (2017), and Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Third Draft 2014 Edition); Guideline for Clinical Trials of New Patent Chinese Medicines (Trial), and Expert Consensus on the Prevention and Treatment for Primary Osteoporosis with Traditional Chinese Medicine (2015)	Random number table	—	Thoracolumbar single-level vertebral body	Comprehensive acupuncture+ zoledronic acid injection + PKP
Kang Wang [50]	2022	China	Guidelines for the Diagnosis and Treatment of Osteoporotic Fractures (2017 Edition) formulated by Chinese Orthopaedic Association	Random number table	2.76 ±0.56 2.87 ±0.60	T10-L3	Internal heating acupuncture + moxibustion + intravenous drip of sodium ibandronate injection combined with calcium carbonate D3 chewable tablets Intravenous drip of sodium ibandronate injection combined with calcium carbonate D3 chewable tablets
Chengwu Geng [51]	2022	China	Professional Committee on Osteoporosis Prevention and Rehabilitation of Chinese Society of Rehabilitation Medicine. Expert Consensus on Diagnosis and Treatment of Osteoporotic Vertebral Compression Fractures (2021 Edition)	Random number table	<2w	T10-L5	Ultra-micro needle knife + PKP
Shun Li [52]	2022	China	Osteoporotic DRF diagnosis formulated in the Journal of Practical Orthopaedics	Random number table	—	Distal radius	Warm needling moxibustion+ small splint fixation + calcium carbonate D3 Small splint fixation + calcium carbonate D3
Cong Huang [53]	2022	China	Expert Consensus on Anti-Osteoporosis Treatment and Management in Patients with Osteoportic Fracture	Random number table	—	Thoracic vertebra	Small needle knife + PVP PVP
Chengjing Liao [54]	2022	China	X-ray	Random number table	—		Micro-needle + celecoxib + PVP Celecoxib + PVP
Wenbing Zhang [55]	2022	China	Guide to Diagnosis and Treatment of Osteoporotic Fracture in China - Osteoporotic Fracture Diagnosis and Treatment Principles (2015)	Random number table	6.04 ± 2. 82 6.13 ± 3.70	T5-L5	Ultrasonic electroacupuncture + routine medication (oral calcium preparations, calcitriol, intramuscular salmon calcitonin); physical therapy: local irradiation of WIRA for 20 min, once every day, oral loxoprofen sodium tablets, tramadol hydrochloride) Routine medication
Jianrong Xie [56]	2021	China	Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Third Draft · 2014 Edition) Osteoporosis, renal deficiency and blood stasis diagnosed based on Guideline for Clinical Trials of New Patent Chinese Medicines (Trial)	Random number table	9.53 ± 5.08 years 9.10 ± 4.93 years	Thoracolumbar spine	Needle knife + Jintiange capsules + routine treatment Routine treatment (calcium carbonate D3 tablets, salmon calcitonin injection, and thermotherapy apparatus used for magnetic thermotherapy)
Xu Chen [57]	2021	China	The Clinical Guideline for Osteoporotic Compression Fractures [J]. Chinese Journal of Osteoporosis, 2015, 21 (06): 643-648 TCM diagnosis meets the diagnostic criteria in the Guideline for Clinical Trials of New Patent Chinese Medicines: Renal deficiency and blood stasis	Random number table	—	Thoracolumbar spine	Warm needling moxibustion + PKP PKP

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Table 1 (*continued*)

First author	Publication period	Author's country	Diagnostic criteria for osteoporotic vertebral compression fractures	Randomization method	Course of disease	Fracture site	Interventional protocol
Fen Zhang [58]	2021	China	Clinical Orthopedics Diagnostics	Random sequence method	—	Thoracolumbar spine	Small needle knife + PVP PVP
Ping Chen [59]	2021	China	Guide to Diagnosis and Treatment of Osteoporotic Fracture in China - Osteoporotic Fracture Diagnosis and Treatment Principles	Random number table	< 3 days	Thoracolumbar spine	Acupuncture + PVP Medication + PVP
Chuanfeng Liu [60]	2021	China	Guidelines for the Diagnosis and Treatment of Primary Osteoporosis (2011)	Random	—	Lumbar vertebra	Floating needle + TCM suffocation TCM suffocation
Xiaoya Liu [61]	2021	China	X-ray	Random number table	—	Lumbar vertebra	Wrist-ankle acupuncture + rehabilitation exercise + PVP and routine postoperative nursing treatment PVP and routine postoperative care
Yating Liu [62]	2021	China	Imaging diagnosis	—	12.02 ±5.02 12.14 ±6.06	T10-L3	Small needle knife + PKP + medication (caltrate, vitamin D, and Modified Liuwei Dihuang Decoction) PKP + medication (caltrate, vitamin D, and Modified Liuwei Dihuang Decoction)
Zhigang Liu [63]	2020	China	Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Third Draft)	Random number table	7.90 ±1.94 8.44 ±1.78	T10-L3	Warm needling moxibustion + PKP + postoperative routine treatment PKP + postoperative routine treatment
Gang Wang [64]	2020	China	X-ray	Random	—	Thoracolumbar spine	Needle knife + routine treatment (calcium carbonate D3 tablets, salmon calcitonin injection, and thermotherapy apparatus used for magnetic thermotherapy) Routine treatment (calcium carbonate D3 tablets, salmon calcitonin injection, and thermotherapy apparatus used for magnetic thermotherapy)
Jianfeng Li [65]	2020	China	—	Random number table	15.52 ± 16.22 16.81 ± 12.31 15.42 ± 9.48	Lumbar vertebra	Floating needle + routine medication (alfacalcidol tablets, calcium and alendronate) + PVP Floating needle + PVP Routine medication (alfacalcidol tablets, calcium and alendronate) + PVP
Yujia Huo [66]	2020	China	Guideline for Clinical Trials of New Patent Chinese Medicines (Trial)	Randomization in the order of seeking treatment	—	Lumbar vertebra	Acupuncture + routine medication (alfacalcidol tablets, caltrate and zoledronic acid injection) Routine medication (alfacalcidol tablets, caltrate and zoledronic acid injection)
Shanshan Chen [67]	2019	China	Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Third Draft · 2014 Edition)	Random number table	4.5±0.7 4.2±0.6	T12, L1, L2	Acupuncture + electroacupuncture + oral medication (salmon calcitonin, calcium carbonate D3) + PKP Oral medication (salmon calcitonin, calcium carbonate D3) + PKP
Zhensheng Chen [68]	2018	China	The Clinical Guideline for Osteoporotic Compression Fractures [J]. Chinese Journal of Osteoporosis, 2015, 21 (6): 643-648	Random number table	—	Thoracolumbar spine	Acupuncture + PVP + oral medication (vitamin D + calcium carbonate tablets) PVP + oral medication (vitamin D + calcium carbonate tablets)
Xiyao Hu [69]	2018	China	Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Third Draft · 2014 Edition)	Random number table	(3.70 ± 1.20) years (3.70 ± 1.20) years	Lumbar vertebra	Jiaji electropuncture + sinomenine hydrochloride injection + medication (elcatonin + VD + calcium carbonate D3) Sinomenine hydrochloride injection + medication (elcatonin + VD + calcium carbonate D3)
Minjian Li [70]	2018	China	The Clinical Guideline for Osteoporotic Compression Fractures [J]. Chinese Journal of Osteoporosis, 2015, 21 (6): 643-648	Random number table	2.39 ±0.28 2.57 ±0.26	Lumbar vertebra	Electropuncture + acupoint injection + PKP surgery and postoperative routine treatment PKP and postoperative routine treatment
Jun Zhao [71]	2017	China	_	Random	25.27 ± 6.861	T10-L3	Small needle knife + external application of traditional Chinese

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Table 1 (*continued*)

First author	Publication period	Author's country	Diagnostic criteria for osteoporotic vertebral compression fractures	Randomization method	Course of disease	Fracture site	Interventional protocol
					25.14 ± 4.256		medicine + PVP + routine medication (calcium carbonate, and Xianlin Gubao capsules) External application of traditional Chinese medicine + PVP + routine medication (calcium carbonate, and Xianlin Gubao capsules)
Meng Pan [72]	2017	China	The Clinical Guideline for Osteoporotic Compression Fractures	Random	—	T10-L5	Acupuncture + Buyang Huanwu Tang + PVP + medication (celecoxib) PVP + medication (celecoxib)
Gang Wang [73]	2017	China	Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Third Draft · 2014 Edition)	Random number table	6.2 ± 3.2 months 6.1 ± 3.3 months	Thoracolumbar spine	Warm needling moxibustion + routine medication (Xianling Gubao and Miacalcic) Routine medication (Xianling Gubao and Miacalcic)
Xiaoqiang Deng [74]	2017	China	_	Random sequence method	_	T9-L4	Small needle knife + PVP PVP
Jinlian Yang [75]	2016	China	Clinical Practice Guidelines for Primary Osteoporosis formulated by the Chinese Medical Association in 2011; The TCM industry standard of the People's Republic of China Criteria of Diagnosis and Therapeutic Effect of Diseases and Syndromes in Traditional Chinese Medicine (1995), diagnosed with renal deficiency and blood stasis syndromes.	Random	7.91 ±4.26 8.25 ±4.68	Thoracolumbar spine	Electropuncture + moxibustion + routine medication (calcitonin and alendronate sodium) Routine medication (calcitonin and alendronate sodium)
Mi Huang [76]	2016	China	Anteroposterior and lateral X-ray film of thoracic vertebra, definitive diagnosis by MRI	Random	—	Thoracic vertebra T5 - T11	Acupuncture + PKP PKP
Wei Zhang [77]	2015	China	Chinese Society of Osteoporosis and Bone Mineral Research. Guidelines for the Diagnosis and Treatment of Osteoporotic Fractures (Discussion Draft) [J]. Chinese Journal of General Practitioners, 2006, (08): 458-459	Random number table	4.3 ± 1.3 4.6 ± 1.6 4.4 ± 1.6	Thoracolumbar spine	Abdominal acupuncture + medication (alfacalcidol, salmon calcitonin, and diclofenac sodium in case of severe pain) Routine acupuncture + medication Medication
Gen Zhang [78]	2013	China	Expert Consensus on the Diagnosis of Osteoporosis in Chinese Population (Second Draft) [J]. Chinese Journal of Osteoporosis, 2000, 6 (1): 1	Random number table	(6.4 ± 2.8) years (7.2 ± 2.8) years	Lumbar vertebra	Acupuncture + routine medication (salmon calcitonin and calcium gluconate) Routine medication (salmon calcitonin and calcium gluconate)
Enliang Su [79]	2003	China	X-ray	Random	<7 days	Hip joint and lumbar vertebra	Acupuncture + routine surgery Medication + routine surgery
Hongmin Deng [80]	2021	China		According to the communication results between doctors and patients	17.52 ±2.51 d	Thoracolumbar spine	PVP Routine medication
Anyi Chen [81]	2020	China	Ct or MRI	——		Thoracolumbar spine	PVP Routine medication
Peide Gao [82]	2017	China	Ct or MRI	Random number table	10.8 ±3.2d 10.2 ±3.5d	Thoracolumbar spine T4 - L5	PKP Routine medication
Yingzhong Wang [83]	2014	China	Ct or MRI	Random		Thoracolumbar spine T10-L5	PVP Routine medication

Intervention measures	Course of treatment	Follow-up or not	Number of case(s)
Dashu, Shenshu, and Zusanli. Matching acupoints: Mingmen and Taixi for kidney deficiency; Sanyinjiao and Pishu for spleen deficiency; Geshu and Sanyinjiao for blood stasis; perform warm needling moxibustion by selecting a 2 cm pure moxa stick to insert it into the needle handle and igniting it near the needle butt after getting the Qi from Shenshu and Zusanli. Retain the needle for 30 min. Twice a week. Retain cupping for 8 to 10 min; perform moving cupping of	3 months	Yes, 6 months after treatment	36 36

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Table 1 (*continued*)

Intervention measures	Course of treatment	Follow-up or not	Number of case(s)
bladder meridian; perform percussopunctator sticking in bladder meridian			
Moxibustion of Shenshu, Zusanli, Guanyuan, Qihai, Mingmen and Zhongwan; ultrasound-guided internal heating acupuncture of five vertebral bodies above and below the lesion site, once a day, 6 days for one course of treatment, followed by rest for a day, for a total of 5 courses of treatment	1 month	Yes, 3 and 6 months after treatment	60 60
Bilateral Shenshu, Qihaishu, Dachangshu, Guanyuanshu, Xiaochangshu, and Ashi at waist; PKP performed regularly at 1, 2 and 4 weeks postoperatively	1 month	Yes, 3 months after treatment	31 30
Hegu, Shousanli, Quchi, bilateral Xuehai, Zusanli, and Guanyuan, with moxa segment inserted into the needle handle, two moxa-cones per acupoint, needle retained for 30 min. Treatment for 5 times every week,	6 weeks	No	34 33
The apex of the superior articular process of the lower vertebral body of fracture treated once every 5 days, for a total of 3 times	15D	Yes, 6 months after treatment	35 28
Hegu micro-needle acupuncture: Routinely disinfect the needling area, select the disposable sterile press needle of Huatuo brand, puncture bilateral Hegu acupoints, and retain the needle for 15 min after needling, twice every day	2W	No	22 22
Ultrasound machine model PHILIPS HD15 Color Doppler ultrasound machine, probe model L5-11, frequency 7.5 MHz; acupuncture needle with a diameter of 0.45 mm and a length of 75 mm for acupuncture; electroacupuncture apparatus: Shanghai G6805-2A electroacupuncture apparatus. Place the patient in the prone position, disinfect with a routine iodophor, spread the surgical drape, apply the coupling agent to the ultrasonic probe, then wear sterile gloves, use iodophor as the coupler between the gloves and the skin, set the probe about 90° to the spine, and use ultrasound to locate the nerve root position (bilateral) above and below the responsible vertebral body, such as T10 and T11 nerve roots located for patients with T11 fractures. Then, under dynamic real-time ultrasound guidance, use the "in-plane" needling method to puncture the needle tip of the acupuncture needle next to the nerve root within the intervertebral foramen (the strong tingling sensation not required). After proper bilateral acupuncture, connect the electroacupuncture apparatus, connect the nerve roots of the same segment on the left and right sides to a pair of positive and negative poles respectively, use dilatational waves, with the intensity being 90% of the maximum intensity that can be tolerated by the patient, and power on for 20 min; once a day, with 7 times for one course of treatment.	1 week	No	23 23
Take the site at which the patient complains of pain and the positive reaction point from the medical examination as the sovereign; the diseased vertebral body as the minister on the left or right side of or below or above the sovereign; other secondary positive points as the assistant, and cervicothoracic, thoracolumbar and lumbosacral junction as the envoy according to the holistic concept and bowstring theory. Use a marker pen to set the point, and perform routine disinfection. b. Hold the needle handle in the right hand, insert the needle knife at the marked point, make the knife-edge line parallel to the longitudinal axis of the vertebral body until the needle knife reaches the bone surface and the corresponding muscle layer, perform the point-cutting release and then remove the needle. c. Apply an adhesive bandage to the postoperative incision, rest for 0.5 h, and allow the patient without discomfort to leave the hospital. Once a week, for a total of 5 times; Jintiange capsules, 3 capsules/time, 3 times every day.	—	Yes, 6 and 12 months after treatment	45 45
Guanyuan, Xuehai, Shenshu, and Hegu, with moxa stick added on the needle, which is retained for 30 min; once every other day	2 months	Yes, 1 year after treatment	35 35
Place the patient in the prone position, and first search the tenderness points near the injured vertebral body, such as thoracolumbar articular process joints, transverse processes, interspinous process spaces, and superior gluteal nerve entrapment points. After marking, perform routine disinfection, spread the surgical drape, subcutaneously inject 1% lidocaine for local infiltration anesthesia at the needling point, select a needle knife of an appropriate length, rapidly penetrate it into the skin according to the treatment standard for the needle knife by the vertical four-step method, then penetrate layer by layer, make the knife-edge line consistent with the direction of muscle fibers, nerves and blood vessels at this site, determine the treatment depth when there is a heavy feeling or crisp sound under the knife edge; at this time, if the patient complains of local acid	1d	Yes, 6 months after treatment	30 30

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Table 1 (*continued*)

Intervention measures	Course of treatment	Follow-up or not	Number of case(s)
swelling pain, the needle knife has reached the pathological reaction point; then, longitudinally strip for 2 to 3 knives, transversely shovel off for 2 to 3 knives, and remove the small needle knife if there is a sense of loosening; use the cotton ball for hemostasis by compression, and apply sterile dressings to the needling point.			
Dashu, bilateral Yaoshu, Mingmen, Yaoyangguan, Zusanli, Sanyinjiao, thoracic vertebral Jiaji, lumbar vertebral Jiaji, Taixi, etc.; Danzhong, Qihai, Neiguan, and Zhigou for chest tightness and palpitation; Tianshu, Guanyuan, Shangwan, and Zhongwan for abdominal distension; Guanyuan, Zhongji, Shuidao, Guilai, and Yongquan for urinary retention; Hegu, Tianshu, Dachangshu, Pishu, and Shangjuxu for constipation; adopt mild reinforcing and attenuating method for acupuncture manipulation, retain the needle for 30 min after getting the Qi of acupuncture, manipulate the needle twice; once a day, 5 days for one course of treatment, rest for 2 days after each course of treatment before moving on to the next one.	1 month	Yes, 3 and 6 months after treatment	63 64
Zhibian, Shenshu, Dachangshu, and local Ashi. Operation method: Place the patient in the prone position, routinely disinfect with iodophor, select a disposable medium-sized floating needle, and let the doctor hold the needle to puncture it at an angle of about 15° to the skin during the operation into the pain point; after the needle body enters the connective tissue layer, perform fan-shaped sweeping movement and inquire the patient's feeling during operation until the patient's pain is significantly relieved. Treat once every other day, 6 times for one course of treatment, for a total of 2 courses of treatment	1 month	No	33 32
Wrist-ankle acupuncture therapy. Divide the patient's body into 6 areas, select the appropriate puncture point on the ankle according to the symptoms, disinfect the skin in the selected area, use a 0.25 mm x 25 mm disposable sterile acupuncture needle of Huatou brand to pierce the skin layer at 30° to the skin, insert the needle along the superficial subcutaneous surface along the longitudinal axis when the patient should feel no aching, tingling and swelling pain, and then fix with the adhesive tape for 2 h; once a day, once every other day after 3 days, 10 days for one course of treatment, for a total of one course of treatment. Guide the patient to do exercise of the lumbodorsal muscle function after wrist-ankle acupuncture every day	10D	No	46 46
Select tenderness points on the spinous process, articular process, transverse process, lumbosacral region, both sides of erector spinae, etc. at the lesion site, mark the needling point, then perform routine disinfection, spread the surgical drape, hold the needle in one hand perpendicular to the skin, make the blade parallel to the longitudinal axis of the body, so as not to damage the nerve vessels and muscle fibers in the corresponding area, penetrate the skin, fascia and soft tissue successively to the bone surface according to the four-step needling method, or when there is a feeling of stiffness under the needle, longitudinally dredge and horizontally strip for 3 - 5 knives until there is a feeling of loosening under the needle, then remove the needle to stop bleeding and wrap with sterile adhesive tape after all the marked tenderness points have been treated. Treatment once a week, once every five days for those with severe pain for one consecutive month			42 42
Shenshu, Taixi, Sanyinjiao, Huantiao, Yanglingquan, Weizhong, Qihaishu, and Jiaji corresponding to the diseased vertebral body. Retain the needle for 30 min; insert a 2 - 3 cm long moxa stick on the needle handle, light it for needle-warm moxibustion, place heat insulation sheet locally, keep the lower end of the moxa stick about 2.5 cm away from the skin, and maintain 20 minutes per acupoint; once every other day for warm needling moxibustion, 5 times every week for 12 consecutive weeks	12 weeks	No	43 43
Observe the patient's low back tenderness points, including the interspinous and supraspinous ligaments, sacroiliac joint and thoracolumbar facet joints, inject an appropriate amount of drugs at local pain points, which are composed of 5 mL of lidocaine and 0.5 mL of vitamin B12 injection. Then insert the needle knife into the needling point, swing it along the transverse axis using the painful tendon and bone strengthening method through lifting, inserting and cutting along the longitudinal axis, and remove the needle knife when there is a sense of loosening under the hand	—	No	40 40

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Table 1 (*continued*)

Intervention measures	Course of treatment	Follow-up or not	Number of case(s)
Assemble the disposable floating needle to the needling device, quickly insert the floating needle into the subcutaneous layer, push forward with the needle tip pointing to the affected muscle until the soft cannula is submerged, and rotate the inner core to the soft cannula. Hold the needle hub by hand, with the needle tip slanted upwards to do the sweeping movement for about 2 min, for a total of about 200 times. During the sweeping movement, perform reperfusion activities of the lower back or buttock muscles, such as abdominal breathing, coughing, little swallow flight, and hip extension.	Three times for one course of treatment, generally for 1 - 3 courses of treatment, continuous treatment each day for the first 3 times, followed by once every 2 to 3 days, for a total of 15 days	Yes, 1, 3, and 6 months after treatment	21 21 19
Sanyinjiao, Zusanli, Yanglingquan, Weizhong, Shenshu, Pishu, Dachangshu, Dashu, Mingmen, Hegu, and Neiguan. Manipulate the needle once in 10 min, retain the needle for 30 min, once a day	2 weeks	No	180 180
Beishu and Jiaji acupuncture combined with electroacupuncture at the lesion site, once a day for six consecutive days, plus one day of rest as one course of treatment, for a total of two courses of treatment.	2 weeks	Yes, 1 week after treatment	36 36
Mingmen, Shenshu, Dazhui, and Dazhu as master acupoints; Xuanzhong, Taixi and Zhaohai added for kidney deficiency; Xuehai, Danzhong and Geshu added for blood stasis. Retain the needle for 30 min, manipulate the needle once every 10 min, and take the mild reinforcing and attenuating method as the main needling method, twice a week.	6 months	No	32 31
Select the pain point on both sides of the spine, use iodophor to disinfect the skin, and select 3 to 6 pairs of Jiaji acupoints according to the patient's condition. During needling, open 0.5 inches at the lower side of the corresponding lumbar spinous processes, direct the needle tip slightly toward the spine direction (about 75° to the skin), penetrate the needle for 20 to 45 mm, and manipulate the needle to get the Qi Then connect the electroacupuncture therapeutic apparatus, set the frequency to 10 Hz, and use dilatational waves to stimulate for 25 to 30 min, with the input intensity of electrostimulation that can be tolerated by the patient. Jiaji electropuncture once a day, six times each week, for a total of three weeks.	3 weeks	No	50 49
Electroacupuncture and acupoint injection treatment, selection of acupoints: Shuangpishu, Weishu, Sanjiaoshu, Shenshu, Qihashu, Dachangshu, Guanyuanshu, Xiaochangshu, Weizhong, and Chengshan.	1 week	Yes, 1 month after treatment	23 23
Perform routine disinfection, spread the surgical drape, apply 1% lidocaine hydrochloride for local anesthesia to the marked point, with 1 mL for each treatment site. If the anesthetic effect is satisfactory, use a No.4 small needle knife to insert the needle perpendicular to the skin along the marked point according to the standard 4-step needling method, keep the knife surface parallel to the muscle fiber direction to deliver it directly to the bone surface while protecting nerves and blood vessels, longitudinally dredge and transversely strip the lesion site for 3 to 4 times, and remove the needle as long as there is a sense of loosening under the needle. After removing the needle, press the needle hole with the sterile cotton ball for 3 to 5 minutes. When there is no bleeding from the needle hole, cover it with an adhesive bandage, and remind the patient to keep the needle hole dry for 3 days. Once a week, determining whether to move on to the next treatment according to the patient's symptoms, for a total course of treatment of 2 weeks.	2 weeks	Yes, 4 weeks after treatment	35 37
Lumbar vertebrae L3 - L5 Jiaji, Zhibian, Huantiao, Tunzhong, Shenshu, Dachangshu, Yinmen, Weizhong, and Chengshan; take tonifying therapy as the needling method, retain the needle for 15 to 20 min, once a day	1 week	No	26 25
Lumbar Jiaji, Ganshu, Pishu, Weishu, Shenshu, Zusanli, Guanyuan, and Qihai. Lumbar Jiaji, Pishu, Weishu, Shenshu, Zusanli, and Guanyuan acupuncture followed by moxibustion, with routine warm needling moxibustion; Ganshu single needling method adopted to get the Qi. Take the finger-twirling supplementation method as the main needling method, with a sense of aching and heaviness in patients, manipulate the needle for 1 min for each acupoint, and retain the needle for 30 min. Once every other day, 15 times for one course of treatment, for a total of 3 consecutive courses of treatment	3 months	No	28 28
First look for the pain point, mark it, then perform routine disinfection, select the model of the needle knife, inject 1% lidocaine subcutaneously at the needling point, adopt the vertical four-step needling method to insert the needle knife into the subcutaneous tissue for incision and dissection. After multi-angle and multi-directional release, remove the small needle knife as long as there is a sense of loosening under the hand; compress the cotton ball to stop	1 day	Yes, 1 week and 1 month after treatment	40 40

(*continued on next page*)

Table 1 (*continued*)

Intervention measures	Course of treatment	Follow-up or not	Number of case(s)
bleeding and use a sterile dressing to protect the needling point. At 24 h postoperatively, wear waist support to carry out off-bed activities for proper exercise of lumbar and back muscle functions, and continue anti-osteoporosis therapy			
Geshu, Pishu, Shenshu, Jiaji (vertebral fracture), Ashi, Weizhong and Yanglingquan. Insert the needle into the above acupoints according to the operation specifications, and connect the G6805-2 electroacupuncture apparatus for stimulation for 30 min after getting the Qi. Then place an ignited moxibustion box on the lower back. Remove the moxibustion box and the filiform needle after 30 min	4 weeks	No	138 135
Jiaji, bilateral Taichong, Neiguan, and Yanglingquan acupuncture; retain the needle for 20 min, once a day, three times for one course of treatment, for a total of two courses of treatment.	1 week	Yes, 1 month, 6 months, 1 year, and 2 years after treatment.	20 20
Abdominal acupuncture: Zhongwan, Xiawan, Qihai, Guanyuan, Shuifen, and Huaroumen; retain the needle for 45 min, once a day, rest for a day after 6 times	3 weeks.	Yes, 1, 2 and 3 weeks after treatment	15 15 15
Take Mingmen, Zhiyang, Waiyangguan, Baihui, and Dazhui as master acupoints; Sanyinjiao, Zusanli, Shangliao, Ciliao, Shenshu, Ganshu, Pishu, Qihai, Guanyuan and Taixi as matching acupoints; needling method: retain the needle for 40 min after getting the Qi of acupuncture, manipulate the needle for 10 min each time, and take tonifying therapy as the main needling method.	15D	1 month after treatment	23 23
Huantiao, Biguan, Yinjiao, Xuehai and other acupoints for hip fracture - femoral intertrochanteric fracture; Shenshu, Pishu, Qihaishu, Weishu and other acupoints as master acupoints for lumbar compression fractures. During expiration, slowly insert a 30G 1.5-inch stainless steel filiform needle into the patient, thrust the needle with force and lift it gently for 1 min after getting the Qi, and then retain the needle for 30 min. During this period, manipulate the needle once, remind the patient to inhale the air to raise the needle speed when removing the needle, and press the needle hole after the needle is removed from the skin. Once every other day, one and a half months for one course of treatment, moving on to the next course of treatment after rest for 10 days, for a total of 2 courses of treatment	3 months	No	25 25
After local anesthesia, place the patient in the prone position, and use X-ray radiography to assist in the puncture via the pedicle approach. After inserting the puncture needle into the anterior 1/3 of the vertebral body, remove the stylet, inject the bone cement, and observe whether there is any leakage; if the situation is satisfactory, remove the puncture needle	Test group: 30 min Control group: 15 days	Yes, 6 months after treatment	44 43
Perform the surgery in the interventional therapy group under the guidance of Imaging instrument, routinely use an 11G or 13G bone puncture needle (COOK, the U.S.A.) to establish the puncture channel, use OSTEOPAL bone cement (Heraeus Medical GmbH, Germany) intraoperatively, and finally use the screw propeller developed by Dragon Crown Medical Co., Ltd. to inject the bone cement.	Test group: 30 min Control group: 3 months	Yes, 1, 3, and 6 months	43 42
Place the patient in the prone position, avoid abdominal compression, perform routine disinfection, and spread the surgical drape. Use two Kirschner wires for cross positioning, clearly position the injury site under the observation with a C-arm machine, and perform invasive anesthesia layer by layer. Insert a percutaneous needle (1.2 mm in diameter) into the pedicle along the injured vertebral body. Insert the puncture needle along the pedicle into the vertebral body and then into the posterior 1/3 of the vertebral body if the positioning is satisfactory. Insert the guide pin into 2/3 of the vertebral body along the inner hole of the puncture needle. The image shows that the lateral puncture needle passes through the central axis of the pedicle and is parallel to the end plate. The AP image shows that the needle tip is 2 cm away from the spinous process connection line and 1.5 cm away from the end plate. Remove the puncture needle, insert the guide pin along the working channel to the anterior 2/3 of the vertebral body, remove the inner core, expand the balloon into the vertebral body, pump the contrast medium, inflate the balloon in the vertebral body, and make clear the status of upper and lower end plates, vertebral body, prevertebral and upper and lower spaces and fracture reduction under the observation with the C-arm machine. Draw the contrast medium, remove the inflated balloon catheter, and slowly inject the bone cement. After about 12 minutes, if the vertebroplasty observed under the C-arm machine is satisfactory and the bone cement is in the vertebral body, sew up the incision and send the patient back to the ward.	Test group: 30 min Control group: 3 months	No	55 55

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Table 1 (*continued*)

Intervention measures	Course of treatment	Follow-up or not	Number of case(s)
Routinely monitor the ECG, oxygen saturation, non-invasive blood pressure, etc. of the patient after entering the room, place the patient in the prone position, with his or her abdomen dangled, determine the projection position of the fractured vertebral body on the body surface through fluoroscopy with the C-arm X-ray machine, and mark the pedicle to be punctured on the skin. Perform routine disinfection, spread the surgical drape, and perform local anesthesia. After incising the skin, insert the needle through the injured vertebral body on the severely compressed side. When inserting the needle into the cortex to half of the pedicle, with the needle tip projected at the middle point of the pedicle, remove the inner core of the needle when the needle is further inserted 2 mm into the posterior margin of the vertebral body (from the pedicle to the anterior 1/3 of the vertebral body), and inject a non-ionic contrast agent through the needle tube for angiography to observe the reflux of paravertebral veins. After confirming there is no mistake through fluoroscopy, slowly inject the bone cement into the vertebral body, and at the same time observe the distribution of bone cement and the changes in the patient's vital signs to avoid unexpected conditions caused by bone cement leakage. When it's confirmed through in vitro observation that the bone cement is basically hardened, the surgery will be completed, and the amount of bone cement injected is generally 4 - 7 mL. Allow the patient to stay in bed for 24 h postoperatively, make a routine X-ray examination to recheck the bone cement in the vertebral body, and apply antibiotics for 1 - 2 days.	Test group: 30 min Control group: 3 months	No	30 38

Gender (male/ female)	General population	Age	Outcome indicators	Bone densitometer
19/17 20/16	72	65.8±7.5 68.23±8.3	VAS/ODI/quality of life score (self-made scale based on QOL and SF-36)/bone density/adverse reaction	
21/39	120	69.93 ± 4.58 70.23 ±5.12	VAS/Pittsburgh Sleep Quality Index (PSQI)/ ODI/ bone mineral density(BMD) index [serum bone alkaline phosphatase and bone gla protein]	Enzyme-linked immunosorbent assay and radioimmunoassay
20/11 18/12	61	73. 2±4. 5 75. 5±6. 0	Evaluation of clinical efficacy according to VAS/ODI/MacNab criteria/COBB angle observed through X-ray radiology	
13/21 11/22	67	62±6 61±5	Fracture healing time, swelling reduction time, pain relief time, Gartland-Werley score, wrist function rating, and distal radius bone density on the unaffected side, detection of serum hypoxia-inducing factor-1α (HIF-1α) and chemerin levels	Bone densitometer measurement
3/32 1/27	73	76.3 ± 7.1 75.3±7.0	VAS/ODI	
9/13 8/14	44	71.24 ±4.08 71.69 ±4.53	VAS/ODI/TCM syndrome score/drug dosage	
7/16 7/16	46	71. 00 ± 7. 56 74. 17 ± 8. 73	Numeric rating scale (NRS)/ODI/serum NO, TNF-α and IL-6	
15/30 14/31	90	70.50 ±9.33 73.39 ±10.04	VAS/ODI/bone metabolism indicators (Type I collagen amino-terminal propeptide, N-terminal osteocalcin, β-collagen degradation product)	
18/17 16/19	70	5 9 .1 ± 5 .6 5 9 .7 ± 5.0	VAS/ODI/COBB angle/serum transforming growth factor TGF-β1 and omentin-1/incidence of fracture after 1 year	
8/22 11/19	60	76.00 ±4.95 74.84 ±6.18	VAS/ODI/JOA	
37/25 36/28	127	57.6±13.2 59.3±12.7	VAS/ADL/SF-36	
11/22 9/23	65	60.82 ±9.65 59.68 ±9.98	VAS/ODI/pain episode duration	
23/23 22/24	92	61.2±2.5 60.9±2.8	VAS/lumbar height/World Health Organization Quality of Life (WHOQOL)	
18/24 17/25	84	73.02 ±7.64	VAS/ODI/bone density (lumbar vertebra L2 - L4, femoral neck, femoral Ward's triangle, hip joint)	Dual-energy X-ray bone densitometer test

(*continued on next page*)

Table 1 (*continued*)

Gender (male/ female)	General population	Age	Outcome indicators	Bone densitometer
		73.33 ±7.16		
30/13 27/16	86	67. 83 ± 7. 04 67. 03± 6. 93	VAS/ODI/COBB angle/blood calcium level/bone density	BMD measurement of the lumbar vertebra anteroposterior position (L2 - L4) and the right calcaneus with Explorer dual-energy X-ray bone densitometer
19/21 20/20	80	73.09± 2.31 73.14± 2.28	VAS	
6/15 5/16 6/13	61	73.48 ± 7.50 75.81 ±6.66	VAS/ODI	
0/180 0/180	360	60.63 ±6.51 61.29 ±7.16	Bone density/bone metabolism indexes (serum calcium (Ca^{2+}), serum phosphorus (P^{3+}) and alkaline phosphatase /quality of life scale (SF-36))	BMD measurement of lumbar vertebra L1 - L5 and femoral neck with EXPERTXL dual-energy bone densitometer (LUNAR, the U.S. A.)
16/20 17/19	72	58.3±6.7 57.6±7.2	VAS/ODI/barthel/QOL	
0/32 0/31	63	76.55 ± 6.78 77.02 ±5.21	VAS/ODI/bone density	BMD measurement of the lumbar vertebra with dual-energy X-ray absorptiometer (GE Lunar Prodigy, the U.S.A.)
18/32 16/33	99	73.90 ±9.14 72.30 ±8.40	VAS/ODI/Interleukin-1β (IL-1β) and osteoprotegerin	
6/17 7/16	46	72.78 ±4.73 71.30 ±4.80	VAS/ODI	
15/20 13/24	72	62.39 ±3.596 62.11 ±3.115	VAS/ODI	
8/18 7/18	51	78.38 ±4.53 78.60 ±3.15	VAS/ODI	
9/19 10/18	56	68±5 68±5	TCM clinical symptom score, BMD, osteocalcin-BGP, and IL-6	BMD measurement of the lumbar vertebra anteroposterior position L2 - L4 and the left greater trochanter (GT) region of the femur before and after treatment with DEXXUMT dual-energy X-ray bone densitometer developed by OsteoSys
21/59	80	61-92	VAS/ODI	
74/66 68/72	273	64±12 66±11	VAS/ Questionnaire Short-Form and patient quality of life score (SF-36)	
3/17 3/17	40	77.7 ± 8.5 (total)	VAS/ODI/COBB angle	
6/9 5/10 7/8	45	65.0±8.4 67.0±6.7 66.6±6.6	VAS/BI	
5/18 6/19	46	67.4±8.28 69.5±7.52	VAS/bone density	BMD measurement of lumbar vertebra L2 - L4 with PA-5 single-photon bone densitometer
18/32 (total)	50	60 to 75 years old	Diet, urine and stool, swelling, local tenderness, callus growth and BMD	XR-36 tester (Norland, the U.S.A.)
24/20 26/17	87	68.16 ±5.27 68.09 ±5.19	VAS/ODI/COBB angle	
9/34 17/25	85	73.8±8.3 74.1±8.4	VAS/ODI/World Health Organization Quality-of-Life Scale (WHOQOL-BREF)	
22/33 20/35	90	70.8± 4.6 70.5±4.5	ODI/NRS/GQOLI-74/COBB angle	
17/13 22/16	68	67.92 ±7.11 66.74 ±8.92	VAS/ODI/COBB angle	

BMD: bone mineral density; BI: Barthel index; GQOLI-74: Generic Quality of Life Inventory; HIF-1α: hypoxia-inducing factor-1α; IL-1β: Interleukin-1β; IL-6: Interleukin-6; JOA: Japanese Orthopaedic Association Scores; NRS: Numeric rating scale; NO: nitric oxide; ODI: Oswestry Disability Index; PVP: percutaneous vertebroplasty; PKP: percutaneous kyphoplasty; PSQI: Pittsburgh Sleep Quality Index; QOL: Quality of Life; SF-36: 36-item Short Form Health Survey; TCM: traditional Chinese medicine; VAS Visual Analogue Scale; WHOQOL: World Health Organization Quality of Life.

O (Outcome): The review primarily examined several outcome indicators, namely the Visual Analogue Scale (VAS), Oswestry Disability Index (ODI), and bone mineral density (BMD). Moreover, secondary outcome indicators, such as the quality-of-life scale, TCM syndrome score, bone metabolism, related inflammatory factor levels, Cobb angle, and the incidence of adverse reactions were also evaluated in the review.

S (Review design): This review only included randomized controlled trials conducted in either Chinese or English.

2.2.2. Exclusion criteria

P (Population): 1) Bone tumors, rheumatoid arthritis, and secondary bone tissue resulting from prolonged oral steroid use, among other factors, can contribute to the development of bone disorders. 2) Individuals with psychosis, dementia, severe neurosis, or similar conditions encounter challenges in assessing treatment efficacy, effectively communicating their condition, and collaborating with healthcare providers. 3) Patients suffering from pathological fractures or metabolic bone disorders are also among those who may potentially derive significant advantages from suitable medical interventions.

I (Intervention): Studies comparing differences in intervention frequency or intensity were excluded from our analysis. Studies that solely investigated the effects of acupuncture at different acupoints were also excluded. The interventions employed in the trial group consisted of traditional Chinese medicine (TCM) and non-drug therapies, excluding acupuncture. These therapies encompassed cupping, scraping, and moxibustion.

C (Comparison): None

O (Outcome): The outcome indicators either remain unknown or are limited to the treatment response rate.

S (Review design): The statistical power of the test could not be met due to the insufficient sample size mentioned in the first statement. In the second statement, it is noted that studies with a cross-over design had a relatively small sample size. The third statement highlights the unavailability of conference abstracts with complete/full text.

2.3. Data sources and search strategy

To identify randomized controlled studies on the treatment of fractures related to osteoporosis using acupuncture, a thorough and systematic search was carried out until April 12, 2023. The search encompassed several reputable databases, namely PubMed, EMBASE, Web of Science, Cochrane Library, China Knowledge Network (CNKI), WanFang, VIP Database, and SinoMed. The search strategy involved employing a combination of subject words and free words, without imposing any geographical limitations.

The specific search strategy is presented in Appendix A.

2.4. Review selection and data extraction

EndNote X9 was utilized to manage and track the studies. The first step in the screening process involved reviewing the titles and abstracts of the articles. This was done to removeeliminate duplicate entries and exclude studies that did not meet the inclusion criteria. Subsequently, the full texts of the remaining studies were carefully examined to select studiesth that met the eligibility criteria. To facilitate the process of data extraction, a standardized data extraction table was created. The extracted data comprises a wide range of aspects. These include the title of the original studies, the primary author's name, the publication year, the author's country of origin, the diagnostic criteria for osteoporosis-related fractures, the method of randomization, the duration of the disease, the site of the fracture, the protocol for intervention, the specific interventions used, the duration of treatment, the follow-up period, the number of cases, the distribution of genders (male/female), the total population included in the studies, the age range, the outcome indicators, and the instrument employed to measure bone mineral density (BMD). The investigators, SQ and QYL, conducted the screening and data extraction procedures independently, ensuring reliability. Thereafter, the accuracy of their work was cross-checked. In case of any discrepancies, a third investigator (WMX) was involved to help resolve them.

2.5. Risk of bias in studies

The Cochrane Collaboration's risk of bias assessment tool [27] was employed by two investigators to evaluate potential bias. This tool comprises seven components, namely: random sequence generation, allocation concealment, blinding of participants and intervention providers, blinding of outcome assessors, incomplete outcome data, selective outcome reporting, and other potential sources of bias. Each element was classified as having a low, high, or unclear risk of bias.

2.6. Outcomes

The efficacy of interventions was primarily assessed in the review by using two outcome indicators: the Visual Analog Scale (VAS) and the Oswestry Disability Index (ODI). The review focused on measuring the efficacy of interventions through these two indicators. Furthermore, the enhancement of Bone Mineral Density (BMD) was regarded as a secondary indicator of outcome. To evaluate the severity of pain reported by the participants, the Visual Analog Scale (VAS) score, which ranges from 0 to 10, was utilized. A score of 0 denoted the mildest pain, while a score of 10 indicated the most intense pain [28]. To assess the functionality of the patients' spines, the ODI, which consists of 10 questions with 6 response options for each question, was utilized. Each response option was assigned a rating from 0 to 5 points, with a higher score indicating a higher level of disability [29]. The measurement of BMD was conducted using dual-energy X-ray absorptiometry, a reliable technique. All three indicators were considered as continuous data. To accommodate variations in baseline values among different studies, the analysis focused on the changes in these indicators from baseline to the end of the studies.

2.7. Synthesis methods

In the network meta-analysis, a Bayesian random effects model was utilized to evaluate the efficacy of various intervention strategies. The modeling procedure employed the Markov Chain Monte Carlo (MCMC) method, with the execution of 4 Markov chains in parallel. The annealing process was repeated 20,000 times, and the modeling achieved convergence after 50,000 simulation iterations. To assess the model fitting and overall consistency, the Deviance Information Criterion (DIC) was employed. The node-splitting method would be utilized to evaluate local consistency in the presence of a closed-loop network. Additionally, intervention measures were compared based on the league table, and the funnel plot was used to accurately represent the heterogeneity among studies. To assess the heterogeneity among studies, a funnel plot was employed. Statistical analysis was conducted using Stata 15.0 (Stata Corporation, College Station, TX) and R4.2.0 (R development Core Team, Vienna, http://www.R-project.org). A significance level of $P < 0.05$ was considered as indicative of a significant difference. Table 1

2.8. GRADE rating

Following the guidelines set forth by the Grading of Recommendations Assessment, Development and Evaluation (GRADE) Working Group [30,31], the evaluation of evidence quality was conducted and the results of the network meta-analysis were presented using the following steps. Initially, the effect sizes and confidence intervals for both direct and indirect comparisons of two intervention measures were analyzed. The GRADE approach was employed to assess the quality of direct comparisons, ensuring consistency with traditional meta-analysis.

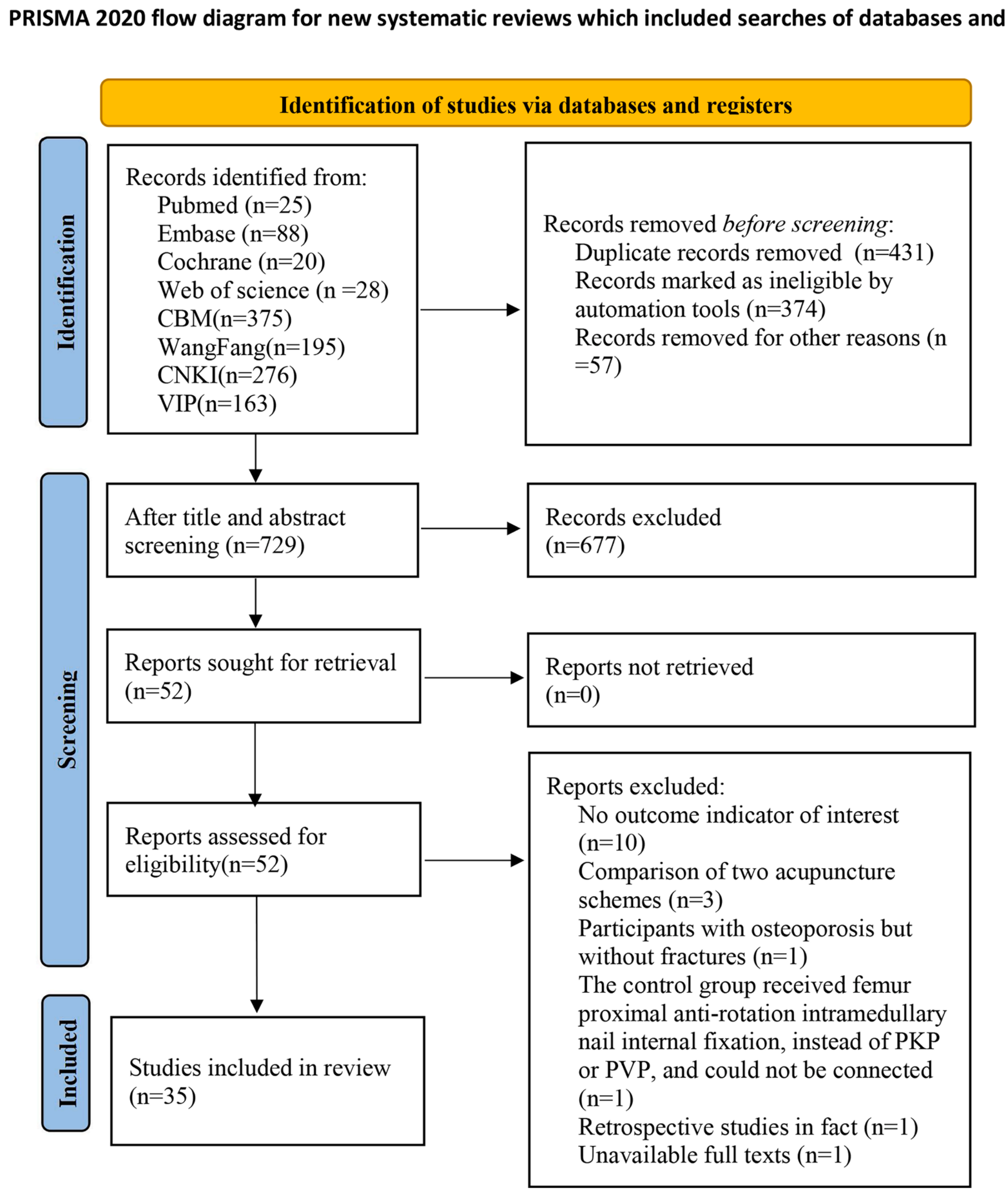

Fig. 1. Diagram of the Studies Selection Process.

Subsequently, the evaluation of indirect evidence was conducted by considering groups with low quality evidence in the direct comparisons, thereby generating indirect results. Finally, the network meta-analysis results were presented and their evidence quality was evaluated.

3. Results

3.1. Study selection

A thorough search was conducted, encompassing a total of 1170 studies. Out of these, 431 duplicate studies were identified and excluded. After carefully evaluating the remaining titles and abstracts, 52 studies were selected for full-text review and downloaded accordingly. Ten articles [32–41] were excluded due to the absence of essential outcome indicators, such as VOS (Visual Outcome Score) and ODI (Oswestry Disability Index). Furthermore, one article [42] was omitted as the original text was unavailable. Additionally, three articles [43–45] were eliminated since they actually compared different acupuncture schemes. Moreover, one article [46] was removed because it solely included individuals with osteoporosis and no fractures. In addition, one article [47] was excluded as the control group underwent proximal femoral anti-rotation intramedullary nail internal fixation surgery instead of PKP (Percutaneous Kyphoplasty) or PVP (Percutaneous Vertebroplasty), rendering it incomparable. Furthermore, one retrospective review [48] was eliminated from the analysis. This systematic review encompasses a total of 35 randomized controlled trials [49–83], comprising a sample size of 2727 subjects. Following rigorous screening and selection, these trials were considered appropriate for inclusion in the analysis (Fig. 1 and Appendix B).

3.2. Study characteristics

In this analysis, a total of 35 studies were included. Among them, 13 studies [50,52,55,56,60,64,66,69,73,75,77–79] reported the combination of daily care with various acupuncture modalities. Conversely, 18 studies [49,51,53,54,57–59,61–63,65,67,68,70–72,74,76] focused on the combination of surgical treatment with different acupuncture modalities for fractures associated with osteoporosis. Specifically, 8 studies

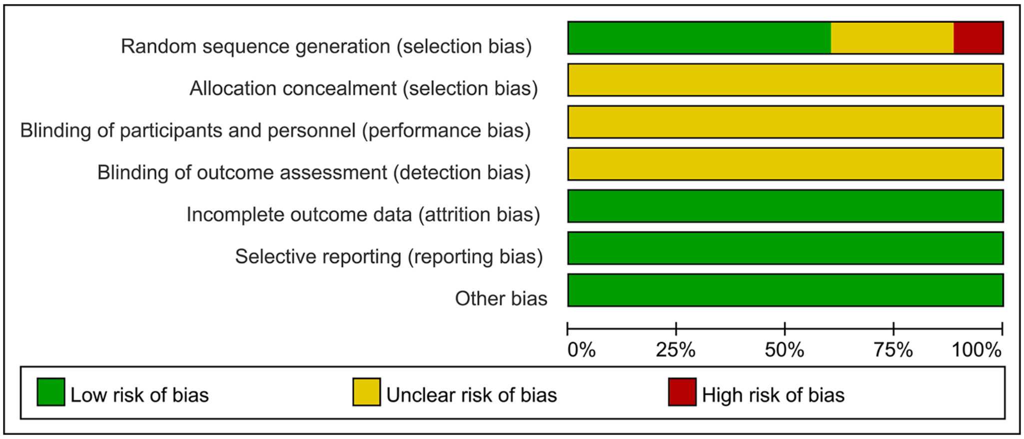

Fig. 2. Results of Assessment of Risk of Bias in the Included Studies.

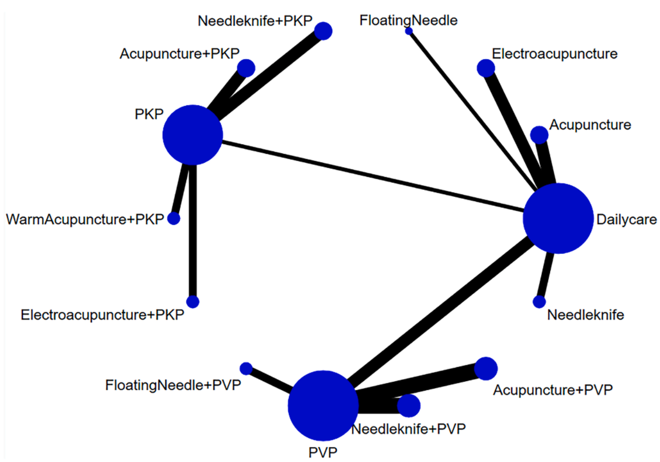

Fig. 3. Network Plot of Pain Relief in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Therapy (Note: (1) The blue dots in the figure represent each intervention method, with the size of the dot indicating the quantity of included studies; (2) The black lines represent direct comparisons between intervention methods, and the thickness of the line corresponds to the amount of studies included in these direct comparisons.).

[53,54,61,65,68,71,72,74] examined the combination of PVP surgery with different acupuncture methods, while 10 studies [49,51,57–59,62,63,67,70,76] investigated the combination of PKP surgery with different acupuncture methods. In total, these studies encompassed five distinct acupuncture modalities. Twelve studies [49,50,54,59,61,66,68,72,76–79] examined the efficacy of ordinary acupuncture, while eight studies [51,53,56,58,62,64,71,74] focused on needle-knife acupuncture. Four studies [52,57,63,73] investigated the effects of warm acupuncture, while electroacupuncture was studied in five studies [55,67,69,70,75]. Two studies [60,65] explored the use of floating acupuncture. Out of the 35 eligible studies, one trial was a three-arm trial [65], and the remaining 34 studies were double-arm trials [49–64,66–83]. The Visual Analog Scale (VAS) score was discussed in 31 studies [49–51,53–65,67–72,74–78,80–83], indicating the level of pain experienced by the participants. The Oswestry Disability Index (ODI) scale was utilized in 24 studies [49–51,53–58,62,63,65,67–72,74,76,80–83], measuring the impact of the treatments on the participants' functional abilities. Additionally, eight studies [49,52,62,63,66,73,78,79] assessed the participants' bone mineral density (BMD) to examine any potential effects on bone health.

3.3. Risk of bias in studies

Concerns about bias have been raised as three of the studies analyzed did not utilize proper randomization methods. However, all studies did provide comprehensive information regarding allocation concealment, blinding procedures, and outcome evaluation. Moreover, there was no

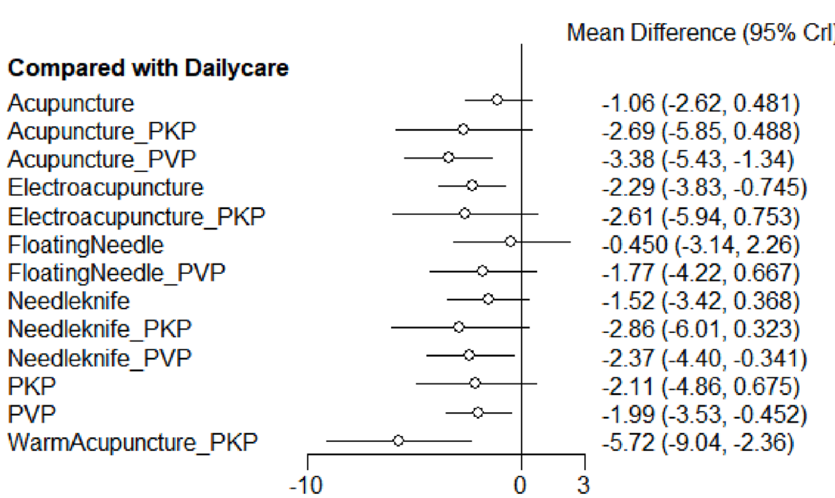

Fig. 4. Forest Plot of Pain Relief in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment.

Table 2
League Table of Pain Relief in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment.

	Acupuncture	Acupuncture_PKP	Acupuncture_PVP	Dailycare	Electroacupuncture	Electroacupuncture_PKP	FloatingNeedle	FloatingNeedle_PVP	Needleknife	Needleknife_PKP	Needleknife_PVP	PKP	PVP	WarmAcupuncture_PKP
Acupuncture	Acupuncture	-1.62 (-5.13, 1.94)	-2.32 (-4.89, 0.25)	1.06 (-0.49, 2.63)	-1.23 (-3.42, 0.97)	-1.53 (-5.21, 2.18)	0.6 (-2.52, 3.74)	-0.71 (-3.61, 2.15)	-0.45 (-2.91, 1.99)	-1.79 (-5.33, 1.77)	-1.31 (-3.88, 1.26)	-1.04 (-4.19, 2.15)	-0.93 (-3.12, 1.26)	-4.65 (-8.33, -0.95)
Acupuncture_PKP	1.62 (-1.94, 5.13)	Acupuncture_PKP	-0.7 (-4.48, 3.08)	2.68 (-0.51, 5.85)	0.39 (-3.18, 3.93)	0.09 (-2.37, 2.54)	2.22 (-1.97, 6.39)	0.9 (-3.12, 4.91)	1.16 (-2.56, 4.86)	-0.17 (-2.36, 2.04)	0.31 (-3.48, 4.08)	0.58 (-0.99, 2.14)	0.68 (-2.87, 4.21)	-3.03 (-5.48, -0.58)
Acupuncture_PVP	2.32 (-0.25, 4.89)	0.7 (-3.08, 4.48)	Acupuncture_PVP	3.38 (1.33, 5.43)	1.09 (-1.48, 3.65)	0.8 (-3.14, 4.72)	2.93 (-0.46, 6.3)	1.6 (-0.73, 3.92)	1.86 (-0.93, 4.66)	0.53 (-3.26, 4.32)	1.01 (-0.88, 2.91)	1.28 (-2.17, 4.73)	1.39 (0.04, 2.73)	-2.33 (-6.26, 1.61)
Dailycare	-1.06 (-2.63, 0.49)	-2.68 (-5.85, 0.51)	-3.38 (-5.43, -1.33)	Dailycare	-2.29 (-3.84, -0.75)	-2.59 (-5.95, 0.76)	-0.46 (-3.16, 2.25)	-1.78 (-4.22, 0.66)	-1.52 (-3.41, 0.38)	-2.85 (-6.02, 0.35)	-2.37 (-4.42, -0.34)	-2.1 (-4.87, 0.69)	-2 (-3.54, -0.45)	-5.71 (-9.07, -2.35)
Electroacupuncture	1.23 (-0.97, 3.42)	-0.39 (-3.93, 3.18)	-1.09 (-3.65, 1.48)	2.29 (0.75, 3.84)	Electroacupuncture	-0.3 (-4.01, 3.41)	1.83 (-1.27, 4.95)	0.51 (-2.37, 3.41)	0.77 (-1.69, 3.21)	-0.56 (-4.1, 3.01)	-0.08 (-2.64, 2.48)	0.2 (-3, 3.39)	0.3 (-1.89, 2.48)	-3.42 (-7.12, 0.27)
Electroacupuncture_PKP	1.53 (-2.18, 5.21)	-0.09 (-2.54, 2.37)	-0.8 (-4.72, 3.14)	2.59 (-0.76, 5.95)	0.3 (-3.41, 4.01)	Electroacupuncture_PKP	2.13 (-2.16, 6.44)	0.81 (-3.35, 4.97)	1.07 (-2.79, 4.92)	-0.26 (-2.73, 2.21)	0.22 (-3.7, 4.13)	0.49 (-1.4, 2.39)	0.59 (-3.1, 4.28)	-3.12 (-5.79, -0.45)
FloatingNeedle	-0.6 (-3.74, 2.52)	-2.22 (-6.39, 1.97)	-2.93 (-6.3, 0.46)	0.46 (-2.25, 3.16)	-1.83 (-4.95, 1.27)	-2.13 (-6.44, 2.16)	FloatingNeedle	-1.33 (-4.96, 2.31)	-1.06 (-4.38, 2.23)	-2.4 (-6.56, 1.8)	-1.91 (-5.29, 1.47)	-1.64 (-5.52, 2.24)	-1.54 (-4.65, 1.57)	-5.25 (-9.57, -0.96)
FloatingNeedle_PVP	0.71 (-2.15, 3.61)	-0.9 (-4.91, 3.12)	-1.6 (-3.92, 0.73)	1.78 (-0.66, 4.22)	-0.51 (-3.41, 2.37)	-0.81 (-4.97, 3.35)	1.33 (-2.31, 4.96)	FloatingNeedle_PVP	0.26 (-2.83, 3.35)	-1.07 (-5.07, 2.97)	-0.59 (-2.9, 1.73)	-0.32 (-4.01, 3.39)	-0.22 (-2.11, 1.68)	-3.93 (-8.1, 0.24)
Needleknife	0.45 (-1.99, 2.91)	-1.16 (-4.86, 2.56)	-1.86 (-4.66, 0.93)	1.52 (-0.38, 3.41)	-0.77 (-3.21, 1.69)	-1.07 (-4.92, 2.79)	1.06 (-2.23, 4.38)	-0.26 (-3.35, 2.83)	Needleknife	-1.33 (-5.04, 2.38)	-0.85 (-3.63, 1.94)	-0.58 (-3.93, 2.78)	-0.47 (-2.92, 1.97)	-4.19 (-8.03, -0.35)
Needleknife_PKP	1.79 (-1.77, 5.33)	0.17 (-2.04, 2.36)	-0.53 (-4.32, 3.26)	2.85 (-0.35, 6.02)	0.56 (-3.01, 4.1)	0.26 (-2.21, 2.73)	2.4 (-1.8, 6.56)	1.07 (-2.97, 5.07)	1.33 (-2.38, 5.04)	Needleknife_PKP	0.48 (-3.3, 4.27)	0.75 (-0.81, 2.31)	0.85 (-2.7, 4.39)	-2.87 (-5.31, -0.41)
Needleknife_PVP	1.31 (-1.26, 3.88)	-0.31 (-4.08, 3.48)	-1.01 (-2.91, 0.88)	2.37 (0.34, 4.42)	0.08 (-2.48, 2.64)	-0.22 (-4.13, 3.7)	1.91 (-1.47, 5.29)	0.59 (-1.73, 2.9)	0.85 (-1.94, 3.63)	-0.48 (-4.27, 3.3)	Needleknife_PVP	0.27 (-3.16, 3.71)	0.38 (-0.96, 1.7)	-3.35 (-7.27, 0.58)
PKP	1.04 (-2.15, 4.19)	-0.58 (-2.14, 0.99)	-1.28 (-4.73, 2.17)	2.1 (-0.69, 4.87)	-0.2 (-3.39, 3)	-0.49 (-2.39, 1.4)	1.64 (-2.24, 5.52)	0.32 (-3.39, 4.01)	0.58 (-2.78, 3.93)	-0.75 (-2.31, 0.81)	-0.27 (-3.71, 3.16)	PKP	0.1 (-3.08, 3.28)	-3.62 (-5.49, -1.74)
PVP	0.93 (-1.26, 3.12)	-0.68 (-4.21, 2.87)	-1.39 (-2.73, -0.04)	2 (0.45, 3.54)	-0.3 (-2.48, 1.89)	-0.59 (-4.28, 3.1)	1.54 (-1.57, 4.65)	0.22 (-1.68, 2.11)	0.47 (-1.97, 2.92)	-0.85 (-4.39, 2.7)	-0.38 (-1.7, 0.96)	-0.1 (-3.28, 3.08)	PVP	-3.72 (-7.4, -0.03)
WarmAcupuncture_PKP	4.65 (0.95, 8.33)	3.03 (0.58, 5.48)	2.33 (-1.61, 6.26)	5.71 (2.35, 9.07)	3.42 (-0.27, 7.12)	3.12 (0.45, 5.79)	5.25 (0.96, 9.57)	3.93 (-0.24, 8.1)	4.19 (0.35, 8.03)	2.87 (0.41, 5.31)	3.35 (-0.58, 7.27)	3.62 (1.74, 5.49)	3.72 (0.03, 7.4)	WarmAcupuncture_PKP

PVP: percutaneous vertebroplasty; PKP: percutaneous kyphoplasty.

indication of selective reporting or significant loss to follow-up in any of the eligible studies. For a concise overview of the assessment results, please refer to Fig. 2.

3.4. Meta-analysis

3.4.1. VAS

VAS is a scale utilized for assessing the pain levels in patients with osteoporotic fractures.

3.4.1.1. Network relationship between interventions. A network analysis of VAS interventions included ordinary acupuncture, electro-acupuncture, floating acupuncture, needle-knife, and warm acupuncture. All of these interventions were used in combination with daily care or surgical treatment. However, there were no direct comparisons made between any two acupuncture methods in these studies. (Fig. 3)

3.4.1.2. Pooled results. The fulfillment of the consistency assumption was confirmed by the discrepancy in DIC coefficients between the consistency model and the inconsistency model. A considerable number of patients chose to undergo active surgical treatment, whereas several patients opted for non-surgical treatment, which involved bed rest combined with drugs and acupuncture treatment. In terms of pain relief, the combination of electroacupuncture and daily care resulted in a significant reduction in the VAS score (VAS= −2.29 (95 % CI: −3.84 ∼ −0.75)), as compared to the use of daily care alone. In addition, the efficacy of other acupuncture methods in relieving pain was not significant. However, when ordinary acupuncture was combined with PVP surgery, there was a notable decrease in the VAS score (VAS= −1.39 (95 % CI: −2.73 ∼ −0.04)), indicating a synergistic effect (refer to Fig. 4 and Table 2). Furthermore, the combination of warm acupuncture with PKP surgery resulted in a significant reduction in the VAS score (VAS= −3.62 (95 % CI: −5.49 ∼ −1.74)).

3.4.1.3. Subgroup analysis. The VAS score was subjected to subgroup analysis, considering the varying follow-up times mentioned in the studies. The impact of this variability in follow-up time was found to be statistically significant, affecting the results of the meta-analysis for the VAS score.

For the 1-month follow-up evaluation, four different acupuncture modalities were utilized: ordinary acupuncture, floating acupuncture, electroacupuncture, and needle-knife. The results of the evaluation clearly demonstrated that electroacupuncture, when combined with conservative treatment, effectively reduced the VAS score (VAS= −2.3, 95 % CI: −3.3 ∼ −1.29), resulting in significant pain relief. Furthermore, when floating acupuncture or ordinary acupuncture was combined with PVP surgery, there was also a significant reduction in the VAS score compared to PVP surgery alone (VAS= −1.39, 95 % CI: −2.67 ∼ −0.09; VAS= −1.31, 95 % CI: −2.34 ∼ −0.28, respectively). However, there were no statistically significant differences in VAS scores observed between PKP surgery alone and PKP surgery in combination with ordinary acupuncture, electroacupuncture, or needle-knife.

Regarding the 6-month follow-up, three different acupuncture modalities were utilized: ordinary acupuncture, floating acupuncture, and needle-knife. When comparing the combination of needle-knife and conservative treatment to receiving only daily care, a significant reduction was observed in the VAS score (VAS= −1.47 (95 % CI: −2.52 ∼ −0.48)). Furthermore, when comparing the combination of ordinary acupuncture with PVP surgery to undergoing PVP surgery alone, a significant reduction was found in the VAS score (VAS= −1.59 (95 % CI: −3.01 ∼ −0.18)). Similarly, when comparing the combination of needle-knife with PKP surgery to undergoing PKP surgery alone, a significant reduction was observed in the VAS score (VAS= −1.26 (95 % CI: −2.28 ∼ −0.24)). Detailed results of the VAS subgroup analysis were presented in Appendix C.

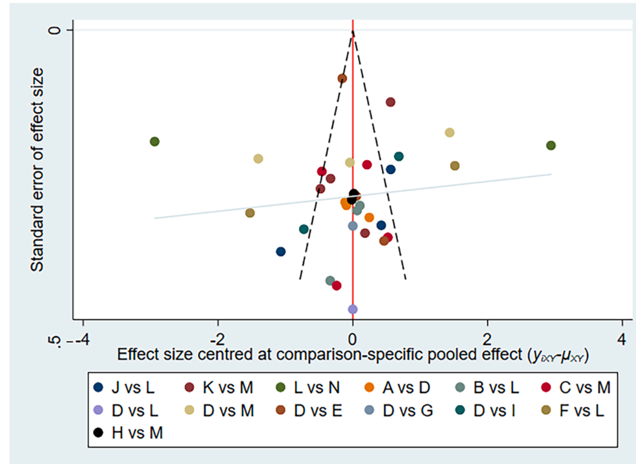

Fig. 5. Funnel Plot of Pain Relief in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment
(Note: In the figure, A-N represent different intervention methods, namely: Acupuncture, Acupuncture+PKP, Acupuncture+PVP, Dailycare, Electro-acupuncture, Electroacupuncture+PKP, FloatingNeedle, FloatingNeedle+PVP, Needleknife, Needleknife+PKP, Needleknife+PVP, PKP, PVP, WarmAcupuncture+PKP).

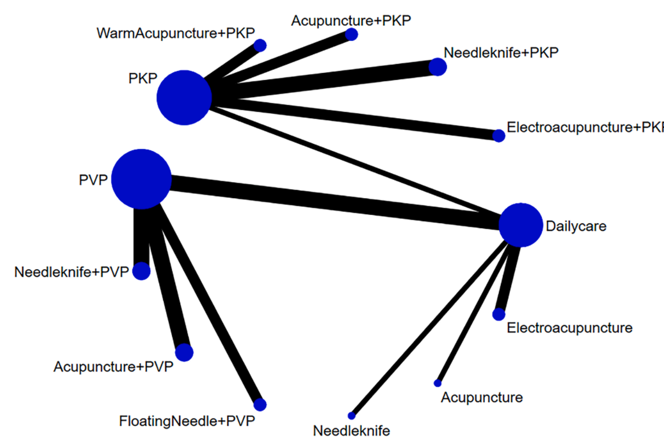

Fig. 6. Network Plot of Disability Improvement in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment
(Note: (1) The blue dots in the figure represent each intervention method, and the larger the dot, the greater the quantity of included studies; (2) The black lines represent direct comparisons between intervention methods, with the thickness of the line corresponding to the amount of studies included in these direct comparisons.).

3.4.1.4. Reporting biases. The funnel plot represents the distribution of the two interventions through different colors. After careful analysis, it was found that none of the studies showed any significant publication bias for VAS (Fig. 5).

3.4.2. ODI

ODI is a scoring scale employed to assess the degree of motor dysfunction in patients with osteoporotic fractures.

3.4.2.1. Network relationship between interventions. The ODI network comprised five interventions, including ordinary acupuncture, electro-acupuncture, floating acupuncture, needle-knife, and warm acupuncture. All of these interventions were administered in combination with daily care or surgical procedures. However, no direct comparisons were conducted to assess the relative efficacy of these acupuncture techniques (Fig. 6).

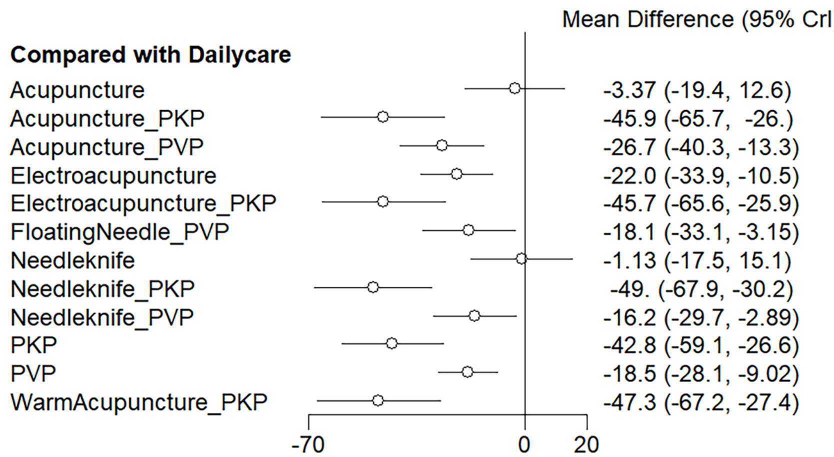

Fig. 7. Forest Plot of Disability Improvement in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment.

3.4.2.2. Pooled results. When comparing the consistency model and the inconsistency model, the differences in DIC coefficients followed the consistency assumption. It is worth noting that the ODI score was significantly reduced (ODI= −22.03 (95 % CI: −33.88 ~ −10.54)) and function improved with the combination of electroacupuncture and daily care. Moreover, when comparing surgical treatment alone with PVP/PKP surgery alone and in combination with various acupuncture methods, there was no significant difference in disability improvement. Fig. 7 and Table 3 provide a visual representation and detailed data of these findings.

3.4.2.3. Subgroup analysis. A subgroup analysis was conducted to specifically examine the influence of follow-up time on the ODI score in studies that reported this outcome measure. The ODI score was found to vary depending on the duration of follow-up, indicating a significant association.

Regarding the 1-month follow-up, a range of acupuncture techniques, such as ordinary acupuncture, electroacupuncture, floating acupuncture, and needle-knife therapy, were employed. An interesting observation was made when comparing the effects of daily care alone to the combination of electroacupuncture and daily care. It was found that the combined approach resulted in a significant decrease in the ODI score. Specifically, the ODI score exhibited a reduction of −22.13 (95 % CI: −36.36 to −8.38). No significant differences in the ODI score were found when comparing PVP surgery alone to PVP surgery combined with ordinary acupuncture, floating acupuncture, or needle-knife therapy. Likewise, there was no notable distinction between PKP surgery alone and PKP surgery combined with ordinary acupuncture, electroacupuncture, or needle-knife therapy.

During the 3-month follow-up period, various acupuncture techniques were implemented, such as ordinary acupuncture, floating acupuncture, needle-knife, and warm acupuncture. It is important to highlight that there were no significant differences in the ODI scores between patients who received only daily care and those who received daily care along with floating acupuncture or needle-knife therapy. No significant differences were found between patients who underwent PVP surgery alone and those who underwent PVP surgery combined with floating acupuncture or needle-knife therapy. Similarly, there were no notable variations in ODI scores when comparing patients who had PKP surgery alone to those who had PKP surgery combined with ordinary acupuncture, needle-knife, or warm acupuncture.

During the 6-month follow-up review, three different acupuncture techniques were utilized: ordinary acupuncture, floating acupuncture, and needle-knife therapy. Our analysis showed that there were no significant differences in ODI (Oswestry Disability Index) scores between daily care alone and a combination of ordinary acupuncture or needle-knife therapy. The comparison between PVP surgery alone and a combination of ordinary acupuncture, floating acupuncture, or needle-knife therapy, as well as the comparison between PKP surgery alone and a combination of ordinary acupuncture or needle-knife therapy, did not yield any significant differences. The ODI subgroup analysis results can be found in Appendix D for a more detailed understanding.

3.4.2.4. Reporting biases. Fig. 8 represents the distribution of the two different interventions using varying colors in the funnel plot. The analysis, as a whole, did not reveal any significant evidence of publication bias for the ODI outcome in each review.

3.4.3. BMD

BMD is an indicator used to measure bone strength.

3.4.3.1. Network relationship between interventions. The BMD network integrated three interventions: standard acupuncture, needle-assisted care, and surgical treatment. Nonetheless, a comparative analysis among these acupuncture techniques was not conducted. (Figs. 9 and 10)

3.4.3.2. Pooled results. Upon comparison of the consistency model and the inconsistency model, it was observed that the differences in DIC coefficients strongly support the consistency assumption. Additionally, when daily care was combined with either ordinary acupuncture or warm acupuncture, there was no significant disparity in BMD. On the other hand, a significant improvement in BMD was found when needle-knife plus PKP surgery (BMD= 0.12 (95 % CI: 0.01~0.22)) was compared to PKP surgery alone. It is worth mentioning that the included studies did not provide any description of PVP surgery combined with different acupuncture methods for improving BMD in Tables 4 and 5.

3.4.3.3. Reporting biases. Unfortunately, the absence of publicly reported studies on BMD prevents us from creating a meaningful funnel plot to illustrate publication bias in the current body of research.

3.5. Grade assessment of evidence level

The studies did not incorporate the intervention methods into a cohesive framework, leading to inconsistency in the reported outcome indicators. Consequently, both direct and indirect comparisons yielded meta-analysis findings and evidence level that were consistent with the observed outcomes. For a comprehensive evaluation of the results, please consult Appendix E.

Table 3
League Table of Disability Improvement in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment.

	Acupuncture	Acupuncture_PKP	Acupuncture_PVP	Dailycare	Electroacupuncture	Electroacupuncture_PKP	FloatingNeedle_PVP	Needleknife	Needleknife_PKP	Needleknife_PVP	PKP	PVP	WarmAcupuncture_PKP
Acupuncture	Acupuncture	-42.45 (-67.95, -16.96)	-23.37 (-44.27, -2.39)	3.37 (-12.58, 19.41)	-18.66 (-38.61, 0.82)	-42.33 (-67.84, -17.05)	-14.68 (-36.55, 7.25)	2.24 (-20.56, 25.08)	-45.61 (-70.39, -20.92)	-12.87 (-33.65, 7.94)	-39.42 (-62.15, -16.6)	-15.16 (-33.78, 3.47)	-43.86 (-69.42, -18.51)
Acupuncture_PKP	42.45 (16.96, 67.95)	Acupuncture_PKP	19.13 (-4.91, 42.93)	45.85 (25.98, 65.74)	23.81 (0.56, 46.65)	0.09 (-16.11, 16.2)	27.8 (2.8, 52.59)	44.75 (19.13, 70.31)	-3.15 (-17.9, 11.61)	29.62 (5.61, 53.35)	3.03 (-8.34, 14.35)	27.31 (5.25, 49.16)	-1.42 (-17.44, 14.72)
Acupuncture_PVP	23.37 (2.39, 44.27)	-19.13 (-42.93, 4.91)	Acupuncture_PVP	26.73 (13.3, 40.29)	4.71 (-13.21, 22.3)	-18.99 (-42.92, 4.97)	8.68 (-6.31, 23.64)	25.61 (4.45, 46.82)	-22.27 (-45.36, 0.86)	10.49 (-2.82, 23.88)	-16.07 (-37.09, 5.13)	8.2 (-1.3, 17.78)	-20.53 (-44.47, 3.37)
Dailycare	-3.37 (-19.41, 12.58)	-45.85 (-65.74, -25.98)	-26.73 (-40.29, -13.3)	Dailycare	-22.03 (-33.88, -10.54)	-45.72 (-65.59, -25.93)	-18.06 (-33.12, -3.15)	-1.13 (-17.48, 15.14)	-48.97 (-67.94, -30.19)	-16.24 (-29.66, -2.89)	-42.79 (-59.11, -26.55)	-18.53 (-28.08, -9.02)	-47.27 (-67.19, -27.44)
Electroacupuncture	18.66 (-0.82, 38.61)	-23.81 (-46.65, -0.56)	-4.71 (-22.3, 13.21)	22.03 (10.54, 33.88)	Electroacupuncture	-23.68 (-46.56, -0.47)	3.99 (-14.88, 23.02)	20.91 (1.06, 41.17)	-26.95 (-48.93, -4.59)	5.8 (-11.71, 23.6)	-20.77 (-40.54, -0.6)	3.53 (-11.35, 18.64)	-25.21 (-48.03, -2.24)
Electroacupuncture_PKP	42.33 (17.05, 67.84)	-0.09 (-16.2, 16.11)	18.99 (-4.97, 42.92)	45.72 (25.93, 65.59)	23.68 (0.47, 46.56)	Electroacupuncture_PKP	27.67 (2.75, 52.59)	44.6 (18.86, 70.24)	-3.27 (-18.06, 11.6)	29.48 (5.65, 53.33)	2.94 (-8.48, 14.4)	27.21 (5.12, 49.25)	-1.55 (-17.59, 14.62)
FloatingNeedle_PVP	14.68 (-7.25, 36.55)	-27.8 (-52.59, -2.8)	-8.68 (-23.64, 6.31)	18.06 (3.15, 33.12)	-3.99 (-23.02, 14.88)	-27.67 (-52.59, -2.75)	FloatingNeedle_PVP	16.93 (-5.32, 39.09)	-30.93 (-55.1, -6.84)	1.82 (-13.05, 16.69)	-24.74 (-46.77, -2.63)	-0.47 (-11.96, 11.08)	-29.2 (-54.08, -4.36)
Needleknife	-2.24 (-25.08, 20.56)	-44.75 (-70.31, -19.13)	-25.61 (-46.82, -4.45)	1.13 (-15.14, 17.48)	-20.91 (-41.17, -1.06)	-44.6 (-70.24, -18.86)	-16.93 (-39.09, 5.32)	Needleknife	-47.86 (-72.75, -23.07)	-15.11 (-36.24, 5.96)	-41.7 (-64.56, -18.7)	-17.41 (-36.28, 1.52)	-46.16 (-71.69, -20.55)
Needleknife_PKP	45.61 (20.92, 70.39)	3.15 (-11.61, 17.9)	22.27 (-0.86, 45.36)	48.97 (30.19, 67.94)	26.95 (4.59, 48.93)	3.27 (-11.6, 18.06)	30.93 (6.84, 55.1)	47.86 (23.07, 72.75)	Needleknife_PKP	32.78 (9.65, 55.77)	6.18 (-3.2, 15.65)	30.46 (9.42, 51.54)	1.72 (-13.03, 16.52)
Needleknife_PVP	12.87 (-7.94, 33.65)	-29.62 (-53.35, -5.61)	-10.49 (-23.88, 2.82)	16.24 (2.89, 29.66)	-5.8 (-23.6, 11.71)	-29.48 (-53.33, -5.65)	-1.82 (-16.69, 13.05)	15.11 (-5.96, 36.24)	-32.78 (-55.77, -9.65)	Needleknife_PVP	-26.56 (-47.47, -5.55)	-2.3 (-11.59, 7.04)	-31.01 (-54.84, -7.12)
PKP	39.42 (16.6, 62.15)	-3.03 (-14.35, 8.34)	16.07 (-5.13, 37.09)	42.79 (26.55, 59.11)	20.77 (0.6, 40.54)	-2.94 (-14.4, 8.48)	24.74 (2.63, 46.77)	41.7 (18.7, 64.56)	-6.18 (-15.65, 3.2)	26.56 (5.55, 47.47)	PKP	24.26 (5.38, 43)	-4.47 (-15.89, 6.92)
PVP	15.16 (-3.47, 33.78)	-27.31 (-49.16, -5.25)	-8.2 (-17.78, 1.3)	18.53 (9.02, 28.08)	-3.53 (-18.64, 11.35)	-27.21 (-49.25, -5.12)	0.47 (-11.08, 11.96)	17.41 (-1.52, 36.28)	-30.46 (-51.54, -9.42)	2.3 (-7.04, 11.59)	-24.26 (-43, -5.38)	PVP	-28.7 (-50.81, -6.79)
WarmAcupuncture_PKP	43.86 (18.51, 69.42)	1.42 (-14.72, 17.44)	20.53 (-3.37, 44.47)	47.27 (27.44, 67.19)	25.21 (2.24, 48.03)	1.55 (-14.62, 17.59)	29.2 (4.36, 54.08)	46.16 (20.55, 71.69)	-1.72 (-16.52, 13.03)	31.01 (7.12, 54.84)	4.47 (-6.92, 15.89)	28.7 (6.79, 50.81)	WarmAcupuncture_PKP

PVP: percutaneous vertebroplasty; PKP: percutaneous kyphoplasty.

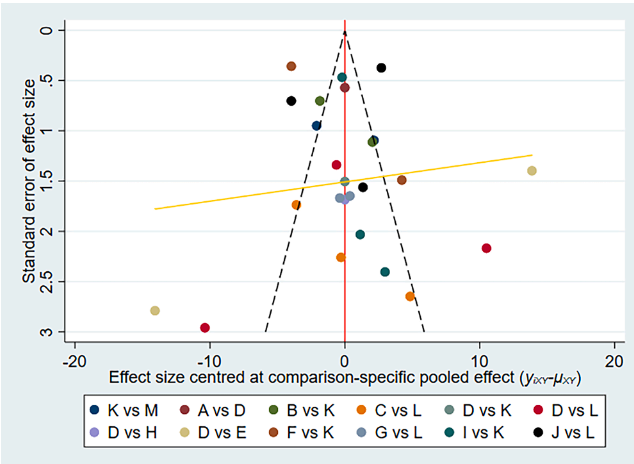

Fig. 8. Funnel Plot of Disability Improvement in Patients with Osteoporosis-related Fractures Using Different Acupuncture Methods as Adjuvant Treatment (Note: In the figure, A-N represent different intervention methods, namely: Acupuncture, Acupuncture+PKP, Acupuncture+PVP, Dailycare, Electro-acupuncture, Electroacupuncture+PKP, FloatingNeedle, FloatingNeedle+PVP, Needleknife, Needleknife+PKP, Needleknife+PVP, PKP, PVP, WarmAcupuncture+PKP).

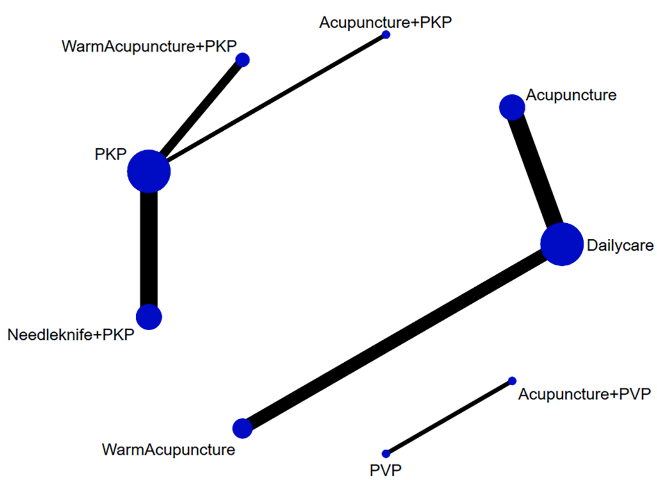

Fig. 9. Forest Plot of BMD Improvement in Patients with Osteoporosis-Related Fractures Receiving Different Acupuncture Methods Combined with Conservative Treatment (Note: (1) The blue dots in the figure represent each intervention method, with the size of the dot indicating the quantity of included studies; (2) The black lines represent direct comparisons between intervention methods, and the thicker the line, the greater the amount of studies included in these direct comparisons.).

4. Discussion

4.1. Summary of the main findings

A combination of daily care and surgical treatment, along with the application of various acupuncture techniques, has demonstrated greater efficacy in alleviating pain and enhancing functional recovery in individuals with osteoporosis-related fractures. Notably, the utilization of electroacupuncture in conjunction with daily care has been observed to expedite the rate of improvement in functional disability. Consequently, electroacupuncture is regarded as a preferred therapeutic approach. It is essential to acknowledge, nevertheless, that these findings are derived from a restricted number of studies and should be approached with careful consideration.

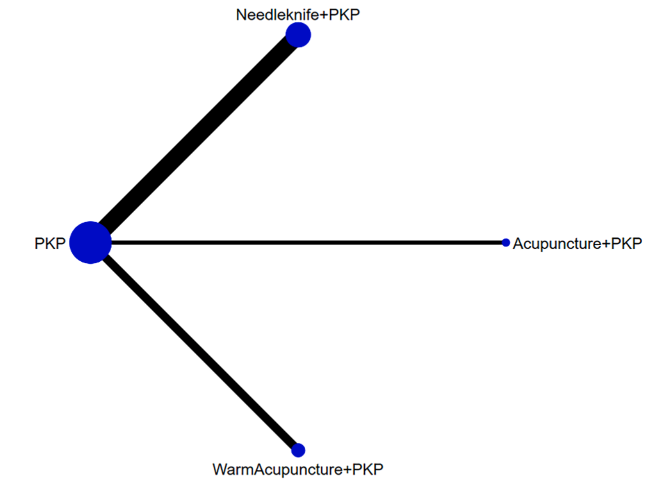

Fig. 10. Forest Plot of BMD Improvement in Patients with Osteoporosis-Related Fractures Receiving Different Acupuncture Methods Combined with Surgery Treatment (Note: (1) The blue dots in the figure represent each intervention method, with the size of the dot indicating the quantity of included studies; (2) The black lines represent direct comparisons between intervention methods, and the thicker the line, the greater the quantity of studies included in these direct comparisons.).

Table 4
League Table of BMD Improvement in Patients with Osteoporosis-related Fractures Receiving Different Acupuncture Methods Combined with Conservative Treatment.

	Acupuncture	Dailycare	WarmAcupuncture
Acupuncture	Acupuncture	-0.15 (-0.38, 0.08)	-0.09 (-0.45, 0.27)
Dailycare	0.15 (-0.08, 0.38)	Dailycare	0.06 (-0.22, 0.34)
WarmAcupuncture	0.09 (-0.27, 0.45)	-0.06 (-0.34, 0.22)	WarmAcupuncture

BMD: bone mineral density

4.2. Comparison with previous reviews

Although the exact causes and mechanisms of osteoporosis are not fully understood, it is known to be the underlying cause of fractures associated with this condition. The management of osteoporotic fractures often includes surgical or conservative treatment options. Surgical intervention is typically recommended for severe vertebral compression fractures [84], while patients who are suitable for conservative treatment may initially undergo conservative management to closely monitor the recovery process [85]. A comparative analysis of the cost-effectiveness of percutaneous vertebroplasty/kyphoplasty (PVP/PKP) surgery and conservative treatment for osteoporosis-related fractures has been conducted in both the United Kingdom and the United States. In the short term, the surgical approach entails higher costs and places a greater burden on medical insurance. However, in the long term, it proves to be more economically efficient in terms of overall quality of life [86,87]. In terms of long-term efficacy, it has been observed that the incidence of refractures is similar among patients with osteoporosis-related fractures who undergo surgical treatment compared to those who receive conservative treatment [14,15].

Addressing the imbalance of bone metabolism is vital in the management of osteoporosis-related fractures. This not only helps prevent refractures but also alleviates pain symptoms and movement disorders associated with these fractures. Research studies have shown that acupuncture can effectively enhance the levels of Bone Mineral Density (BMD) in individuals with primary osteoporosis and reduce the scores on the Visual Analog Scale (VAS) [88]. Furthermore, the utilization of

Table 5
League Table of BMD Improvement in Patients with Osteoporosis-related Fractures Receiving Different Acupuncture Methods Combined with PKP Surgery.

	Acupuncture_PKP	Needleknife_PKP	PKP	WarmAcupuncture_PKP
Acupuncture_PKP	Acupuncture_PKP	0.08 (-0.19, 0.35)	-0.04 (-0.29, 0.21)	-0.01 (-0.3, 0.28)
Needleknife_PKP	-0.08 (-0.35, 0.19)	Needleknife_PKP	-0.12 (-0.22, -0.01)	-0.09 (-0.27, 0.1)
PKP	0.04 (-0.21, 0.29)	0.12 (0.01, 0.22)	PKP	0.03 (-0.12, 0.18)
WarmAcupuncture_PKP	0.01 (-0.28, 0.3)	0.09 (-0.1, 0.27)	-0.03 (-0.18, 0.12)	WarmAcupuncture_PKP

BMD: bone mineral density; PVP: percutaneous vertebroplasty; PKP: percutaneous kyphoplasty.

acupuncture as a treatment for patients with fractures has demonstrated its efficacy in suppressing the harmful effects of free radicals and regulating the activity of antioxidant enzymes during the healing process. This leads to a reduction in the negative impacts caused by oxygen free radicals, ultimately aiding in the promotion of fracture healing [89]. Moreover, in cases of thoracolumbar compression fractures caused by osteoporosis, a combination of conventional acupuncture with PVP (Percutaneous Vertebroplasty) or PKP (Percutaneous Kyphoplasty) has been proven to effectively alleviate pain, improve bone mineral density (BMD), and enhance overall quality of life [90].

4.3. Strengths and limitations

The limitations of this review should be acknowledged, despite its significance as a reference for future clinical applications. It comprehensively examined the efficacy of diversified acupuncture in treating fractures associated with osteoporosis. However, it is crucial to consider the constraints and potential biases that may have influenced the findings. 1) Acupuncture, a traditional Chinese medicine treatment, has gained global recognition and is increasingly being used in combination with surgery or conservative treatment for osteoporotic fractures. However, there is a limited number of studies on this topic, and most of them are published in Chinese. Nevertheless, the incorporation of more traditional Chinese treatment methods in clinical practice could be beneficial, as demonstrated by the positive therapeutic effects of traditional Chinese medicine in the treatment of the COVID-19 pandemic. 2) It is important to note that the scarcity of certain interventions in the studies analyzed is a common limitation in current network meta-analyses. Therefore, the results of the network meta-analysis should be interpreted cautiously. However, it is crucial not to disregard the clinical effects of these treatment methods, as a few studies have highlighted their contributions to clinical treatment. 3) Despite the majority of studies being conducted in China, there was minimal correlation between populations and investigators, which helps reduce publication bias. 4) The limited number of studies investigating the Visual Analog Scale (VAS) score at the 3-month follow-up may potentially influence the analysis results. Consequently, a detailed description of the subgroup analysis for the VAS score at the 3-month follow-up was not provided. 5) During the systematic search and comprehensive screening process, it was observed that no individual review specifically focused on the use of acupuncture alone in the treatment of osteoporosis-related fractures. This is because current clinical guidelines recommend fundamental conservative treatments for osteoporosis-related fractures, such as oral calcium supplementation, pain management according to the situation, minimizing physical activities, and bed rest. 6) The efficacy of acupuncture and moxibustion may vary depending on the acupoints used, but determining the efficacy of specific acupoints is challenging given the current body of evidence in evidence-based medicine. Consequently, there is a lack of studies solely focusing on acupuncture as a standalone treatment for osteoporosis-related fractures, making it difficult to determine its therapeutic effects. 7) To compare the advantages of different acupuncture ways in conservative and surgical treatments together, four studies that compared percutaneous kyphoplasty (PKP) and vertebroplasty (PVP) were included.

5. Conclusions

The utilization of acupuncture treatment in a rational and comprehensive manner can effectively alleviate the pain and disability experienced by patients with fractures related to osteoporosis, irrespective of the option of surgical intervention. While there are various acupuncture regimens available, electroacupuncture has demonstrated a superior efficacy in this context. However, it is crucial to include diverse ethnic groups in future studies to validate its efficacy, given that acupuncture is deeply rooted in traditional Chinese medicine.

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CRediT authorship contribution statement

Ruya Sheng: Writing – original draft, Resources, Funding acquisition, Conceptualization. **Mingxia Wu:** Methodology, Investigation, Formal analysis. **Yali Qiu:** Methodology, Investigation, Formal analysis. **Qing Sun:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current review.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.eujim.2024.102378.

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